

PinchPilot Project Proposal

Small hand movements for webcam desktop control

NUS-ISS Pattern Recognition Systems Practice Module

Practice Module Group 15

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Thumb and middle finger move the pointer; thumb and index finger click

Planned one-hand interaction with forearm support and small movements

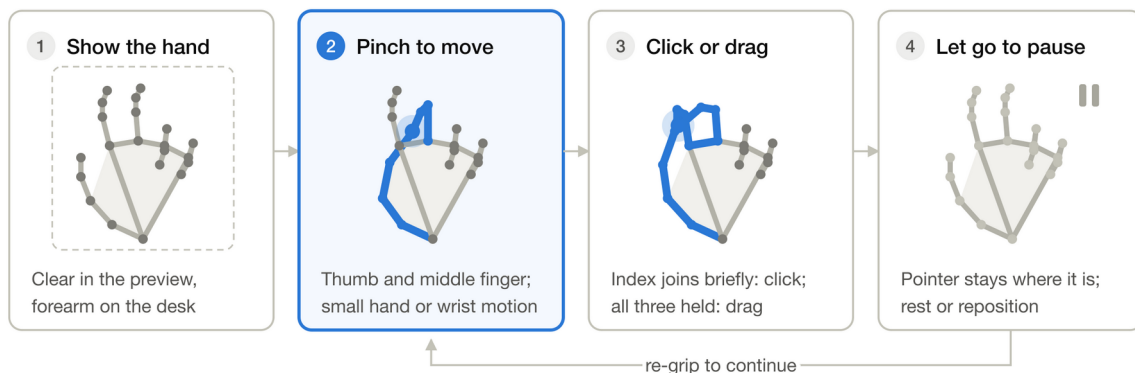


Figure 1 Planned user interaction flow

Introduction

PinchPilot will let laptop users point, click, drag and scroll with an ordinary webcam and small hand or wrist movements. It will prioritise stable control with a supported forearm and a hand that remains visible to the camera, on macOS and Windows.

Initial users will be students and everyday laptop users seeking short periods of touch-free control while reading, browsing or presenting at a desk. A mouse or trackpad will remain available alongside it.

Project objectives

We will build the interaction and desktop tool, collect a gesture dataset, compare rules with learned contact models, and evaluate stability, errors and effort. The planned outputs are a working product and reproducible evidence from a small usability pilot [1].

1 Project Background and Market Context

Example scenario. A student is reading lecture notes on a laptop at a desk. They want to follow a link, scroll to the next section and return to reading without repeatedly reaching for the mouse. PinchPilot will aim to support this sequence with a supported forearm, small movements and an easy pause when the hand needs to rest.

The problem we will address

A webcam can recognise a hand without making that hand a comfortable pointing device. Large movements can tire the arm, small corrections can be overwhelmed by tracking jitter, and making a click can move the pointer away from the target. Research on mid-air interaction identifies arm fatigue as a design concern and motivates measuring effort directly [2].

PinchPilot will focus on the everyday gap between a gesture demo and a tool that users can comfortably return to. Users should be able to make a small correction, click intentionally, pause to rest and resume without an unexpected cursor jump.

Initial users and situations

Situation	What the user needs
Reading and browsing	Scroll a document or webpage and select a link without repeatedly reaching for a mouse.
Presenting at a desk	Point, select and navigate content while remaining in view of the laptop camera.
Brief alternative input	Use a different interaction style for a few minutes and return easily to the mouse or trackpad.

Product opportunity

Pawvis provides webcam-based pointing, clicking, dragging and scrolling on macOS, with configurable gestures and trained gesture templates [3]. It will be a useful product reference. PinchPilot will concentrate on an explicit thumb-middle movement grip, a separate thumb-index click, natural hand/wrist adjustments, and a shared interaction design on macOS and Windows.

Our proposed difference will be this combination of control choices and its evaluation in supported, low-motion use. We will measure the benefit rather than assume that a different gesture is automatically more comfortable or more accurate.

2 User Needs and Product Features

The main interaction will use one control hand and the primary screen. Movement will be engaged deliberately, while separate finger contacts will trigger mouse actions. This will help distinguish an intentional command from a hand that is simply visible.

Function	Proposed gesture	Design purpose
Point	Hold thumb and middle finger together, then move the hand or wrist slightly.	Small movements with deliberate control.
Click and double click	Touch thumb and index finger, then release; repeat for a double click.	Active clicks with less target displacement.
Right click	Briefly pinch thumb and ring finger.	A distinct action for context menus.
Drag	Keep the thumb-middle grip and add the index to the thumb. After a short hold, move; release the index to drop.	A familiar hold, move and release sequence.
Scroll	Form a V with index and middle fingers and move slightly up or down.	Read long pages while holding the pointer still.
Pause and reposition	Release the thumb-middle grip, reposition and re-grip.	Rest or return inside the camera view.

A typical interaction

1. Enable camera control and place the supported hand where the preview shows it clearly.
2. Pinch thumb and middle finger to move the pointer onto a link or object.
3. Briefly pinch thumb and index to click. To drag, hold thumb, index and middle together, wait briefly, then move; release the index to drop.
4. Release the movement grip to rest or reposition; re-grip to continue.

After positioning, the user will also be able to release the middle finger and make a thumb-index click at the held pointer position. This will provide an alternative when maintaining both gestures feels awkward. Hand translation and wrist adjustments will share one control mode.

Control panel and everyday use

A compact preview, adjustable sensitivity, gesture switches and a practice area will support the main workflow. Users will see whether the system is following the hand, holding a click or paused, with Esc and a visible stop control always available.

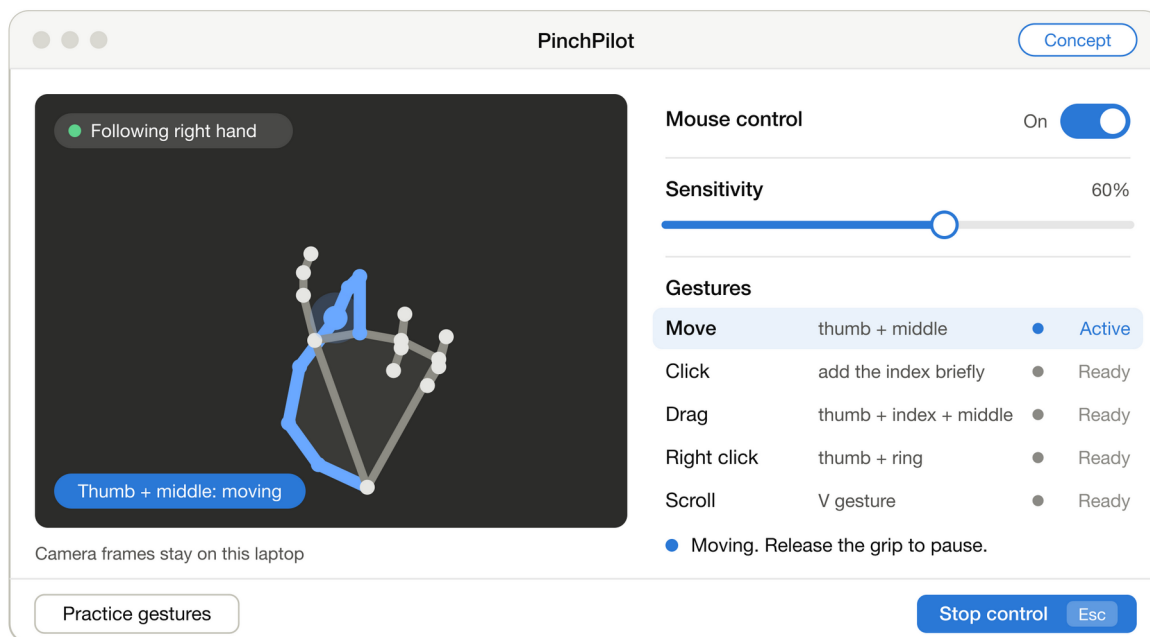


Figure 2 Concept interface with preview, sensitivity and control state

Getting started

Setup will explain camera and mouse-control permissions, show a camera-visible resting position and introduce the gestures. Users will practise pointing, clicking, dragging and scrolling in a dedicated area before controlling other applications. The preview will identify the active hand and the current action.

Comfort and recovery

Continuous pointer sensitivity and scroll-speed settings will let users find a comfortable movement range. Optional calibration will use visible guidance and sufficient valid samples. Boundary hints will help keep the hand in view; releasing a gesture will allow rest or repositioning. Tracking loss will pause control and release held buttons before a fresh movement reference is established.

3 Market Research and Competitive Analysis

The table compares documented roles with PinchPilot’s proposed focus. It is a positioning comparison; the project will use user tasks to establish performance.

Existing option	Main role	PinchPilot focus
Mouse and trackpad	Familiar direct desktop input.	Offer a complementary touch-free option for brief tasks.
Pawvis [3]	macOS webcam control with configurable and trained gestures.	Combine explicit pinch roles, small hand/wrist motion and macOS/Windows support.
MediaPipe [4, 5]	Hand-tracking building block for applications.	Build a usable interaction, contact decision and desktop experience around the tracking.

Pawvis will serve as a product reference [3]. If included in matched tasks, the comparison will focus on pointer response, displacement during clicks and control stability with a relaxed hand. The tested version, camera setup and control settings will be recorded.

4 Project Scope and System Design

Hand keypoints drive two decisions: where to point and which fingers touch

Planned processing on every camera frame, all on the laptop

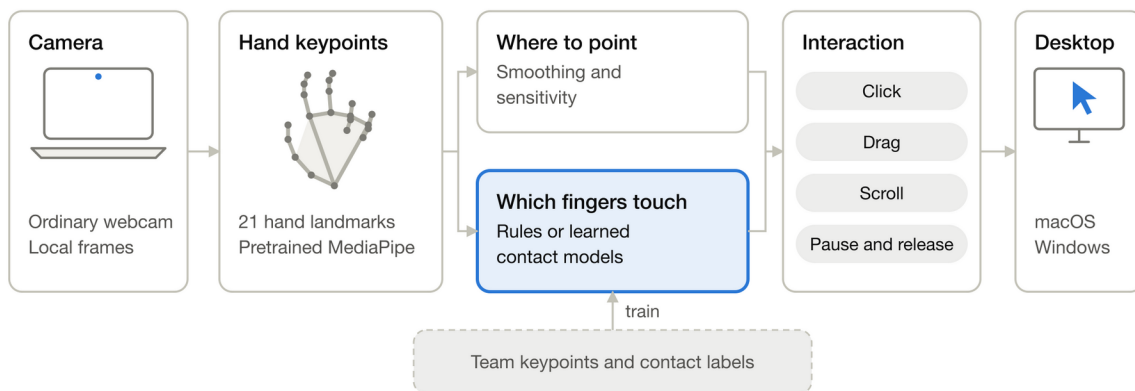


Figure 3 Proposed processing flow

A Python and PySide6 application will share one control engine across macOS and Windows. Hand movement will drive the pointer; recognised contacts will drive clicks and dragging. A common interaction layer will coordinate these actions, scrolling and pause/recovery. This will let recognisers be compared without changing the product’s gesture design.

The initial scope will be one control hand, the primary screen and short, seated use. Guidance, continuous sensitivity adjustment and optional calibration will help users find a comfortable camera-visible position. The system will lock one hand, pause on tracking loss and release held buttons. Normal operation will process camera frames locally.

5 Pattern Recognition Techniques

The learning task will be deliberately focused: identify thumb-middle, thumb-index and thumb-ring contact from hand-keypoint sequences. MediaPipe will supply the pretrained tracking [4, 5]. We will compare a random forest [7] and a small temporal CNN [8] with geometric rules, while smoothing will balance pointer jitter and responsiveness [6].

Technique area	Role in the product
Supervised learning / unsupervised learning scenarios	Supervised: reviewed contact labels will train the recognisers. Unsupervised: an autoencoder will learn features from unlabelled keypoint sequences by reconstruction.
Machine learning / Deep learning techniques	Compare a random forest and a small temporal CNN for contact recognition.
Hybrid machine learning / Ensemble approach	Random-forest tree ensemble; autoencoder pretraining followed by supervised fine-tuning; learned contact decisions with rule-based interaction and recovery.
Intelligent sensing / sense making techniques	Turn camera observations into hand keypoints, hand states, stable movement and desktop actions.

The module requires at least three of these four areas [1]; the table states our planned contribution in each. MediaPipe itself will not be retrained. Personal model adaptation will remain an extension after the general controller is evaluated, with tuned rules available as a fallback.

A second-priority experiment will study denoising-autoencoder pretraining for contact recognition [10], comparing the same CNN with and without pretraining. Its product question is whether varied unlabelled hand use improves difficult-posture recognition or reduces the amount of contact labelling needed.

6 Data Sources and Collection

The main dataset will be recorded by the three team members for the proposed gestures. Each person will complete at least three sessions on different days, about 15 minutes each, using their laptop camera. Sessions will vary supported posture, hand orientation, distance and lighting; camera and participant effects will be reported as a limitation of this small dataset.

Source	Material and use
Team recordings	Contact/release, short taps, grip-plus-click and near-contact negatives; keypoints and reviewed labels for training and recognition tests.
Unlabelled usage records	Team members may explicitly enable keypoint-only recording during ordinary tasks to capture varied poses for pretraining; no video is saved.
User tasks	Matched click, drag and scroll tasks; action records and feedback for evaluating the product.
Synthetic examples	Controlled trajectories for software and regression checks only.

7 Data Preparation and Model Training

The data process will turn recorded examples into a recogniser that can be assessed inside the product. Each step will retain enough information to explain errors and reproduce the comparisons.

7.1 Collect and organise examples

Recordings will retain timestamps, hand keypoints, pseudonymous member IDs, session IDs and camera conditions. Guided examples will cover the three contact pairs, combinations of movement and click gestures, and actions that should not trigger a command. Missing-hand periods will remain visible in the records.

7.2 Review contact labels

The team will label whether each finger pair is touching and mark uncertain moments separately. Prompts will help organise trials rather than serve as unquestioned ground truth. With consent, short hand-only reference clips will support label review; a second member will check a sample of events. Clips will be deleted after review, and only permitted keypoints and labels will be shared.

7.3 Train and integrate the recogniser

The random forest will provide a classical ML comparison; the temporal CNN will learn patterns across a short sequence of hand observations [7, 8]. Candidate models will first be compared with rules on recordings, then checked alongside the live controller before being allowed to issue mouse actions. The selected recogniser will share the same gesture behaviour and pause/recovery handling.

The same causal CNN encoder will also be pretrained to reconstruct masked joints or parts of a recent keypoint sequence, then fine-tuned on contact labels [10]. Masking will also cover derived features that reveal a masked target. Classification will retain the three fingertip-distance features. The study will assess contact recognition; reconstructed coordinates will not drive the pointer.

7.4 Evaluate on separate people and sessions

Recognition will use three-fold leave-one-person-out evaluation: one member's complete recordings are held out for testing, while the other two provide training and separate validation sessions. Pretraining uses only the training partition, excluding unlabelled validation/test recordings too. Model selection and settings are frozen before each test; results will be reported per person and summarised.

Label-efficiency comparisons will use approximately 25%, 50% and all available training labels, sampled as complete recordings or gesture episodes, with fixed validation/test sets and the same unlabelled pool. Results will be grouped by hand orientation and partial occlusion where enough valid examples exist.

Optional personal adaptation may use an earlier unlabelled session of the test person, then compare adapted and unchanged models on the same later session. Adaptation needs no new contact labels; evaluation still needs reviewed labels. This condition will be reported separately. Synthetic results will remain software checks.

8 Evaluation Plan and Expected Deliverables

Matched comparisons will assess contact recognition, pointer stability and task usability, separating the effects of learned recognition and control refinement.

Area	Baseline or comparison	Main measures
Contact recognition	RF/CNN versus tuned geometric rules on the same held-out recordings.	Precision, recall, F1; missed/false events; unavailable input.
Autoencoder pretraining	Same CNN with/without pretraining at three label budgets.	F1, missed/false events and latency; posture-group results.
Pointer control	Basic mapping versus stabilised hand/wrist control; same recogniser.	Jitter, click displacement, movement, re-grips and corrections.
User tasks	Basic rules; stabilised rules; stabilised control with learned recognition.	Completion, time, errors, effort, NASA-TLX [9] and preference.
Product reference	Mouse/trackpad; optional Pawvis on the same Mac [3].	Task performance and experience across products.

Internal usability pilot

Three members will complete matched pointing, clicking, dragging and scrolling tasks with models trained and selected without their own recordings. Condition order will rotate, with equal practice and forearm support. Results, failures and feedback will be reported per person. Three developer-testers provide an internal pilot, not evidence of general-population usability.

Work plan

The proposal is due on 30 September, the first presentation is on 6 October, and final submission is on 31 October [1].

Period	Planned work	Checkpoint
30 Sep–6 Oct	Design gestures; build webcam tracking, pointing, click, drag, right click, scroll, pause, preview and settings; pilot the recording protocol.	Basic desktop demo and recording protocol.
7–16 Oct	Refine hand/wrist control, sensitivity, calibration and recovery; build practice/recording tools; collect and review team data.	Refined controller and labelled recordings.
12–23 Oct	Compare rules, RF and CNN; add autoencoder pretraining as second priority. Validate and freeze before held-out tests.	Recognition results; configurations frozen by 23 Oct.
24–27 Oct	Run the internal usability pilot in three conditions with frozen models and settings.	Per-person task results and feedback.
24–31 Oct	Analyse findings; package macOS/Windows builds; complete guidance, report, slides, video and individual submissions.	Final submission on 31 Oct.

Development and data work will overlap. The first presentation will report actual progress against the planned demo and recording protocol. Each recognition configuration will be frozen before its held-out test; the formal usability pilot will start only after configurations are fixed on 23 October.

Tuned rules remain the fallback. If data collection slips, autoencoder experiments will be deferred first; personal adaptation and Pawvis tasks follow core evaluation.

Expected deliverables

Delivery will include macOS/Windows builds, user guidance, the consented dataset, code/models, reproducible evaluations, report, slides and a 10–15 minute video, plus individual reports and peer review [1].

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AI assistance statement

OpenAI Codex and Anthropic Claude assisted with source research, proposal drafting, translation, review, figure design and document layout. The team will review the final content and references and remains responsible for their quality. AI assistance used in software development and experiments will be recorded separately.